

PRACTICE QUESTIONS

1. An error 2% in excess is made while measuring the side of a square. The percentage of error in the calculated area of the square is:
2. The shortest distance between two opposite corners of a rectangular park is 68 metres, while the ratio of the length to the breadth of the park is 15 : 8, respectively. Find the difference between the cost of fencing boundary of the park at the rate of Rs. 12/metre and cost of sodding the park at the rate of Rs. 2/metre².
3. A and B can do a piece of work in 72 days. B and C can do it in 120 days, A and C can do it in 90 days. In how many days all the three together can do the work?
4. A can do a work in 6 days and B in 9 days. How many days will both take together to complete the work
5. The ratio of the breadth to the length of a rectangular field is 1 : 3. The total cost of cutting the grass of the field at the rate of Rs 6 per m² is Rs. 8712. Find the cost of fencing the boundary of the field at the rate of Rs. 11 per m
6. The breadth of a rectangular field is 80% of its length. Area of square is six times of breadth of the rectangle. If the ratio of Area of Rectangle to Perimeter of Square is 15 : 1, then what will be the length of rectangle?
7. A and B together can do a piece of work in 10 days. A alone can do it in 30 days. The time in which B alone can do it is
8. A alone can complete a work in 12 days. A and B together can complete it in 8 days. How long will B alone take to complete the work?
9. A and B can complete a piece of work in 8 days, B and C can do it in 12 days, C and A can do it in 8 days. A, B and C together can complete it in
10. The age of Abhi, 7 years from now, will be 25% more than the age of Sunny 5 years hence. 5 years ago the average of age of Abhi's father and Abhi was 23 years. Abhi was 4 years old when Sunny was 1 year old. What will be the age of Abhi's father when Sunny was 10 years old?
11. Average weight of group of four persons P, Q, R and S is 67 Kg. If Q is excluded from the group and another person T is included in the group then average weight decreases by 4 kg. If the average weight of T and P is 58 kg and weight of T is 18 kg more than the weight of P, then find the average weight of Q and T.
12. Aman goes to park daily. His last week average speed with which he completed one round of the park was 47Km/h, for the first four days was 37Km/h, and that for the last four days was 52.5Km/h. Find out the time taken by Aman to travel 203 Km if he travels with the speed of the fourth day.
13. The average of 15 numbers is 26. If each number is first multiplied by 6 then increased by 6 then find the new average.

14. The average height of the first six students is 170 cm, the average height of the last eight students is 175 cm. The average height of the total 16 students is 180 cm. Find the average height of the rest two students.

15. Average temperature from Sunday to Wednesday is 34°C while average temperature from Wednesday to Saturday is 38°C . Average temperature throughout the week is 36°C . Find the temperature on Wednesday.

16. The average of the marks obtained in an examination by 8 students was 51 and by 9 other students was 68. The average marks of all 17 students was :

17. The average of 30 results is 20 and the average of other 20 results is 30. What is the average of all the results?

18. The average of 10 numbers is 7. If each number is multiplied by 12, then the average of the new set of numbers will be

19. The average weight of five persons sitting in a boat is 38 kg. The average weight of the boat and the persons sitting in the boat is 52kg. What is the weight of the boat?

20. 12 kg of rice costing 30 per kg is mixed with 8 kg of rice costing 40 per kg. The average per kg price of mixed rice is

21. Two pipes A and B can fill a tank in 10 hours and 15 hours respectively while a third pipe C can empty the full tank in 20 hours. All the pipes are opened for 5 hours and then C is closed. Find the time in which the tank is full?

22. Two pipes A and B can fill a tank in 8 minutes and 12 minutes respectively. If both the pipes are opened simultaneously, after what time should B be closed so that the tank is full in 6 minutes?

23. Two pipes can separately fill the tank in 15hrs and 30hrs respectively. Both the pipe are opened and when the tank is $\frac{1}{3}$ full a leak is developed due to which $\frac{1}{3}$ water supplied by the pipe leaks out. What is the total time to fill the tank?

24. Three pipes A, B and C is attached to a cistern. A can fill it in 20 minutes and B can fill it in 30 minutes. C is a waste pipe. After opening both the pipes A and B, Riya leaves the cistern to fill and returns when the cistern is supposed to be filled. But she found that waste pipe C had been left open, she closes it and now the cistern takes 5 minutes more to fill. In how much time the pipe C can empty the full cistern?

25. Two pipes A and B can fill a tank in 20 minutes and 30 minutes respectively. If both pipes are opened together, the time taken to fill the tank is :

26. A tap can empty a tank in one hour. A second tap can empty it in 30 minutes. If both the taps operate simultaneously, how much time is needed to empty the tank?

27. Two pipes can fill a cistern in 3 hours and 4 hours respectively and a waste pipe can empty it in 2 hours. If all the three pipes are kept open, then the cistern will be filled in :

28. 12 pumps working 6 hours a day can empty a completely filled reservoir in 15 days. How many such pumps working 9 hours a day will empty the same reservoir in 12 days ?

29. Three taps A, B, C can fill an overhead tank in 4, 6 and 12 hours respectively. How long would the three taps take to fill the tank if all of them are opened together?
30. Next term of the sequence 508 53 212 29 60 ?
31. Next term of the sequence 12 6 15 67.5 ? 3729.375
32. Next term of the sequence 132 156 182 210 240 ?
33. Next term of the sequence 188 269 333 382 418 ?
34. The next number of the sequence 3, 5, 9, 17, 33 is
35. Next term of the sequence 8, 12, 9, 13, 10, 14,, is
36. In the sequence of number 0, 7, 26, 63,, 215, 342 the missing term is
37. The value of x in the sequence 1, 2, 6, 24, x is
38. Which number in the sequence 8, 27, 64, 100, 125, 216, 343 is wrongly written?
39. I. $5x + 9y = 109$ II. $3x + 4y = 62$
40. I. $2x^2 - 24x + 70 = 0$ II. $y^2 - 20y + 91 = 0$
41. If $a * b = 2a - 3b + ab$, then $3 * 5 + 5 * 3$ is equal to
42. If $a * b = 2(a + b)$, then $5 * 2$ is equal to :
43. If x is real, then the minimum value of $(x^2 - x + 1)$ is
44. If a and b be positive integers such that $a^2 - b^2 = 19$, then the value of a is
45. If $a * b = a^b$, then the value of $5 * 3$ is
46. 59.71% of $(207.93 + 218.09) = (?)^2$
47. $17.92 \times 10.97 - 52.11 = ?$
48. $101.68 \div 5.93 = ? \div 11.01$
49. $48.88 \times 3.06 + 70.09\%$ of $60.23 - 11.986 \times 2.919 + ? = 58.263$
50. What approximate value will come in place of the question mark (?) in the following question? 14.98% of $239.99 = 19.99\%$ of ? - 25.02% of 199.99
51. What approximate value will come in place of the question mark (?) in the following question?
 $(800.01\%$ of $74.99) / 59.98 = ?^2 - \sqrt{675.99}$
52. What will come in the place of question mark (?) in the given expression?
 $[{(1458 \div 9) \div 6} \div 3] + 16^2 + \sqrt{225} = ?$
53. What approximate value will come in place of the question mark (?) in the following question? 44.98% of $600 + 35.89 \times 4 - 396.12 \div 17.95 = ?$

54. What approximate value will come in place of the question mark (?) in the following question?

$$? = 29.91\% \text{ of } (35.2 \times (144.01 + 5.95))$$

$$55. ? = 24.95 \times (559.83 \div 13.91) - 40.01\% \text{ of } 779.79 + 6.01^2$$

56. From a bank, Ram and Shyam together took a certain amount under simple interest and they lent the total amount to Mohan at 2% more simple interest. At the end of 4 years, the total money earned by Ram after paying the interest to the bank was Rs. 400 more than that of Shyam. From the bank, the total amount taken by Ram was how much more than that of Shyam?

57. The salary of a man is Rs. 60000, from which he deposits x% amount at 13% simple interest. If the accumulated amount for the sum deposited after 3 years was Rs. 29190, then find the value of 'x'.

58. The rate of interest for the first 2 years is 3% per annum, for the next 3 years is 8% per annum, and for the period beyond 5 years is 10% per annum. If the man withdraws total amount of Rs. 5320 after 6 years, find the sum he deposited?

59. Sanjay borrowed certain amount of money at simple interest at the rate of 5% p.a. for the first three years, 10% p.a. for the next five years and 12% p.a. for the period beyond 8 years. If the total interest paid by him at the end of 12 years is Rs. 6780, how much money did he borrow?

60. A certain sum 10M Invested at simple Interest becomes 130M in 30 years. If the same amount invested with same rate of Interest but at Compound Interest, it will become ___ after two years?

61. At what percent per annum will Rs. 3000/- amounts to Rs. 3993/ in 3 years if the interest is compounded annually?

62. The compound interest on Rs. 8,000 at 15% per annum for 2 years 4 months, compounded annually is:

63. The principal, which will amount to Rs. 270.40 in 2 years at the rate of 4% per annum compound interest, is

64. The compound interest on Rs. 30,000 at 7% per annum for a certain time is Rs. 4,347. The time is

65. Rs. 800 at 5% per annum compounded annually will amount to Rs. 882 in?

66. The rate of simple interest for which Rs. 6,000 will amount to Rs. 6,900 in 3 years is?

67. A sum amounts to double in 8 years by simple interest. Then the rate of simple interest per annum is

68. The simple interest on a sum for 5 years is one fourth of the sum. The rate of interest per annum is

69. A sum of money amounts to Rs. 850 in 3 years and to Rs. 925 in 4 years at some rate of simple interest. The sum is
70. The population of a village decreases at the rate of 20% per annum. If its population 2 years ago was 10,000, the present population is
71. In the income statement of Asha and Ravenna, the ratio of their income in the year 2017 was 5 : 4. The ratio of Asha's income in the year 2018 to that in 2017 is 3 : 5 and the ratio of Ravenna's income in the year 2018 to that in 2017 is 3 : 2. If Rs. 10242 is the sum of the income of Asha and Ravenna in the year 2018, then find the income of Ravenna in the year 2017?
72. The average score in an examination taken by 52 students of a class is 85. If the scores of the best 5 performers are not considered, the average score of the remaining students falls by 2. If, none of the first five highest scorers is not below 80 and if each of the 5 top scorers had distinct integral scores, find the maximum possible score of the topper
73. Incomes of John and Kelvin are in the ratio 4 : 7 and their spending are in the ratio 6 : 11. If John saves one third of his income, then what will be the ratio of their savings
74. Three numbers A, B and C are in the ratio of 12 : 15 : 25. If sum of these numbers is 312, ratio between the difference of B and A and the difference of C and B is –
75. Out of three positive numbers, the ratio of the first and the second numbers is 3 : 4 that of the second and the third numbers is 5 : 6 if the product of the second and the third numbers is 4320. What is the sum of three numbers?
76. If $a : b = 7 : 9$ and $b : c = 15 : 7$, then what is $a : c$?
77. If $A : B = 3 : 4$, $B : C = 5 : 7$ and $C : D = 8 : 9$ then $A : D$ is equal to
78. If $a : b = 5 : 7$ and $c : d = 2a : 3b$, then $ac : bd$ is :
79. If $a : b = 5 : 7$ and $c : d = 2a : 3b$, then $ac : bd$ is :
80. On mixing two classes A and B of students having average marks 25 and 40 respectively, the over all average obtained is 30. Find the ratio of the students in the class A and B.
81. Marks of two candidates P and Q are in the ratio of 2 : 5. If the marks of P is 120, marks of Q will be
82. A bag contains 8 apple and 6 orange. Four fruits are drawn out one by one and not replaced. What is the probability that they are alternatively of different fruits?
83. A box contains tickets numbered from 1 to 16. 3 tickets are to be chosen to give 3 prizes. What is the probability that at least 2 tickets contain a number which is multiple of 4?
84. A basket contains 8 Red and 6 Pink toys. There is another basket which contains 7 Red and 8 Pink toys. One toy is to drawn from either of the two baskets. What is the probability of drawing a Pink toys?
85. 12 persons are seated around a round table. What is the probability that two particular persons sit together?

86. Three numbers are in the ratio 5 : 6 : 7. If the sum of their squares is 990, then the numbers are
87. Two numbers are in ratio of 21 : 26. If 8 is added in each, the new numbers are in ratio of 5 : 6. Find the ratio of numbers, if 6 is subtracted from each number?
88. The income of Asaram, Satpal and Rahim in the ratio of 12 : 9 : 7 and their spendings are in the ratio 15 : 9 : 8. If Asaram saves 25% of his income. What is the ratio of the savings of Asaram, Satpal and Rahim?
89. The ratio of zinc and copper in a brass pieces is 13 : 7. How much zinc will be there in 100 kg of such a piece?
90. A shopkeeper deals in milk and 45 litre mixture is to be distributed in Milk & Water in the ratio of 4 : 1. If 4 litre milk & 3 litre water will be added in the mixture then what will be the new ratio of water and milk?
91. 48 litre of Glycerin is mixed with 144 litre Rose water. D litre of total mixture is taken out and 32 litre Glycerin and 48 litre Rose water are added in the mixture. The final mixture contains 30% Glycerin, find the quantity of the mixture that is taken out.
92. In a mixture, the ratio of the alchohol and water is 6 : 5. When 22 litre mixture are replaced by water, the ratio becomes 9 : 13. Find the quantity of water after replacement
93. Copper and Zinc are in the ratio 2 : 3 in 200 gms of an alloy. The quantity (in grams) of copper to be added to it to make the ratio 3 : 2 is:
94. The ratio of milk to water in three containers of equal capacity is 3 : 2, 7 : 3 and 11 : 4 respectively. The three containers are mixed together. What is the ratio of milk to water after mixing?
95. How many litres of water should be added to a 35-litre mixture of milk and water containing milk and water in the ratio of 4 : 3 such that the resultant mixture has 60% water in it?
96. A milk vendor has 21 litres of cow milk, 42 litres of toned milk and 63 litres of double toned milk. If he wants to pack them in cans so that each can contains same litres of milk and does not want to mix any two kinds of milk in a can, then the least number of cans required is
97. If 42 persons consume 144 kg of wheat in 15 days, then in how many days will 30 persons consume 45 kg of wheat?
98. A can do a piece of work in 12 days and B can do it in 18 days. They work together for 2 days and then A leaves. How long will B take to finish the remaining work?
99. Ramesh and Rahman can do a work in 20 and 25 days respectively. After doing collectively for 10 days at the work, they leave the work due to illness and Suresh completes rest of the work in 3 days. How many days Suresh alone can take to complete the whole work ?
100. A shopkeeper purchases an article for Rs. 3,550 and spends Rs. 50 on it for its repair. If he then sold the article for Rs. 3,816, the percent of profit is

101. Sohan bought an old Honda Bike and spent Rs. 1500 on its repairs. Then Sohan sold it to Rakesh at a profit of 20%. Rakesh sold it to Raj at a loss of 10%. Raj finally sold it for Rs. 12100 at a profit of 10%. How much did Sohan pay for the old Honda Bike?

102. The cost prices of three sports items, Ball Jacket, Bat Socks and Thigh Pad Shoes, are in the ratio 2 : 3 : 4 respectively. If these three items are sold such that a profit of 20% is registered on Jacket, a profit of 25% is registered on Socks and a loss of 10% is incurred on Shoes, then which of the following gives the overall percentage of profit/loss made in the three transactions put together?

103. A shopkeeper marked a product, 20% above the cost price and sold the product for Rs. 3888 by giving two successive discounts of 10% each. Find the cost price of the product and the loss percentage of the shopkeeper.

104. The marked price of a John Players Denim is Rs. 1490. On the occasion of Rakshabandhan, the showroom offered two successive discounts of 11.25% and D% respectively. If an individual purchased the Denim for Rs. 1193.44, then find the value of 'D'.

105. Cost price and marked price of an article is in ratio 4:5. After allowing a discount of Rs. 100, ratio between marked price and selling price is 10:9. Find the ratio of profit after discount as a percent of cost price to the profit without discount as a percent of selling price?

106. A man sold 20 apples for Rs. 100 and gained 20%. How many apples did he buy for 100?

107. A person bought 50 pens for Rs. 50 each. He sold 40 of them at a loss of 5%. He wants to gain 10% on the whole. Then his gain percent on the remaining pens should be?

108. A radio is sold at a profit of 20%. Had it been sold for Rs. 60 more the profit would have been 30%. The cost price of the radio is?

109. A tradesman marks his goods at 20% above the cost price. He allows his customers a discount of 8% on marked price. Find out his profit percent.

110. A shopkeeper marks his goods 20% above cost price, but allows 30% discount for cash. His net loss is : ?

111. The perimeters of five squares are 24 cm, 32 cm, 40 cm, 76 cm and 80 cm respectively. The perimeter of another square equal in area to sum of the areas of these squares is :

112. The ratio of the area of a square to that of the square drawn on its diagonal is

113. The difference between the length and breadth of a rectangle is 23 m. If its perimeter is 206 m, then its area is

114. A street of width 10 metres surrounds from outside a rectangular garden whose measurement is 200 m $\times$ 180 m. The area of the path (in square metres) is?

115. If the length and breadth of a rectangle are in the ratio 3 : 2 and its perimeter is 20 cm, then the area of the rectangle (in cm^2) is

116. The side QR of an equilateral triangle PQR is produced to the point S in such a way that QR = RS and P is joined to S. Then the measure of $\angle PSR$ is?

117. In $\triangle ABC$, the line parallel to BC intersects AB and AC at P and Q respectively. If $AB : AP = 5 : 3$, then $AQ : QC$ is :

118. ABC is an isosceles triangle with $AB = AC$. A circle through B touching AC at the middle point intersects AB at P. Then $AP : AB$ is :

119. In an isosceles triangle, if the unequal angle is twice the sum of the equal angles, then each equal angle is

120. A bag contains 4 white, 5 red and 6 blue balls. Three balls are drawn at random from the bag. The probability that all of them are red, is:

121. Four persons are chosen at random from a group of 3 men, 2 women, and 4 children. The chance that exactly 2 of them are children, is:

122. At an ice cream store, there are 5 flavors of ice cream: strawberry, vanilla, chocolate, mint, and banana. How many different 3-flavor ice cream cones can be made?

123. In how many different ways can the letters of the word 'LEADING' be arranged in such a way that the vowels always come together?

124. A man goes 24 m due west and then 10 m due north. Then the distance of him from the starting point is

125. The LCM of two numbers is 864 and their HCF is 144. If one of the number is 288, the other number is :

126. LCM of two numbers is 225 and their HCF is 5. If one number is 25, the other number will be

127. Product of two co-prime numbers is 117. Then their L.C.M. is

128. The product of two numbers is 2160 and their HCF is 12. Number of such possible pairs is

129. The least number which when divided by 4, 6, 8, 12 and 16 leaves a remainder of 2 in each case is

130. What is the measure of central angle of the arc whose length is 11 cm and radius of the circle is 14 cm

131. A kite is flying at a height of 50 metre. If the length of string is 100 metre then the inclination of string to the horizontal ground in degree measure is

132. A man 6 ft tall casts a shadow 4 ft long, at the same time when a flag pole casts a shadow 50 ft long. The height of the flag pole is

133. A vertical stick 12 cm long casts a shadow 8 cm long on the ground. At the same time, a tower casts a shadow 40 m long on the ground. The height of the tower is

134. A tower standing on a horizontal plane subtends a certain angle at a point 160 m apart from the foot of the tower. On advancing 100 m towards it, the tower is found to subtend an angle twice as before. The height of the tower is
135. Three persons A , B, and C complete a piece of work in 6 days for which they are paid a sum of Rs. 480. If the efficiency of A, B and C are in ratio 4 : 5 : 7, then find the daily income of B?
136. Rashmi and Pallavi can make a carpet in 3 days and 12 days more than the time taken if both of them worked together. Find the time in which Rashmi can make the carpet alone.
137. A group of men decided to do a job in 4 days, but 20 men dropped out everyday. Find the number of men who initially decided to do the job, if job was completed in 7 days?
138. Raj can do a piece of work in 20 days and Rohan can do it in 12 days. On which date will they complete the work, if they work together on prime number dates starting on 29th April?
139. An athlete runs 200 metres race in 24 seconds. His speed (in km/ hr) is :
140. The speed of a bus is 72 km/hr. The distance covered by the bus in 5 seconds is
141. You arrive at your school 5 minutes late if you walk with a speed of 4 km/h, but you arrive 10 minutes before the scheduled time if you walk with a speed of 5 km/h. The distance of your school from your house (in km) is
142. A and B travel the same distance at speed of 9 km/hr and 10 km/ hr respectively. If A takes 36 minutes more than B, the distance travelled by each is
143. A man is walking at a speed of 10 kmph. After every km, he takes a rest for 5 minutes. How much time will he take to cover a distance of 5 km?
144. A single discount equivalent to the successive discounts of 10%, 20% and 25% is
145. A dinner set is quoted for Rs. 1500. A customer pays Rs. 1173 for it. If the customer got a series of two discounts and the rate of first discount is 15% then the rate of second discount was
146. A tap is dripping at a constant rate into a container. The level (L cm) of the water in the container is given by the equation $L = 2 - 2^t$, where t is time taken in hours. Then the level of water in the container at the start is
147. If the product of first fifty positive consecutive integers be divisible by 7^n , where n is an integer, then the largest possible value of n is
148. A man is born in the year 1896 A.D. If in the year x^2 A.D. his age is $x - 4$, the value of x is
149. If m and n ($n > 1$) are whole numbers such that $m^n = 121$, the value of $(m - 1)^{n+1}$ is
150. If $\sqrt{3}^n = 2187$ then the value of n is :
151. Four bells ring at the intervals of 5, 6, 8 and 9 seconds. All the bells ring simultaneously at some time. They will again ring simultaneously after
152. The greatest number, which when subtracted from 5834, gives a number exactly divisible by each of 20, 28, 32 and 35, is

153. If 5 men and 5 women work together then they can finish a work in 5 days but if 5 women work alone then they take $\frac{40}{3}$ more days than the time required by 5 men. Find efficiency of one woman is how much percentage less than one man?

154. A trader sells his goods at 20% profit. Had he bought it at 10% more and sold it for Rs. 70 more, he would have earned a profit of 25%. Find the cost price of the goods.

155. A trader sold two bullocks for Rs. 8,400 each, neither losing nor gaining in total. If he sold one of the bullocks at a gain of 20%, then the other is sold at a loss of A.

156. After getting two successive discounts Shalini got a shirt at Rs. 136 whose marked price is Rs. 200. If the second discount is 15% find the first discount.

157. The owner of a television shop charges his customer 16% more than the cost price. If a customer paid Rs. 16588 for a television set, then what was the cost price of the television set?

158. By selling 32 oranges for Rs. 30 a man losses 25%. How many oranges should be sold for Rs. 24 so as to gain 20% in the transaction?

159. A vender sells lemons at the rate of 5 for Rs. 14, gaining thereby 40%. For how much did he buy a dozen lemon?

160. A dealer buys an article marked at Rs. 25000 with 20% and 5% off. He spends Rs. 2000 on its repair and sells it for Rs. 25,000. What is his gain or loss percent?

161. A sold a watch to B at a gain of 20% and B sold it to C at a loss of 10%. If C bought the watch for Rs. 216, at what price did A purchase it?

162. A man buys a cycle for Rs. 1400 and sells it at a loss of 15%. What is the selling price of the cycle?

163. On selling an article for Rs. 651, there is a loss of 7%. The cost price of that article is

164. A man bought an old typewriter for Rs. 1200 and spent Rs. 200 on its repair. He sold it for Rs. 1680. His profit percent is

165. A fan is listed at Rs. 150 and a discount of 20% is given. Then the selling price is

166. By selling a tape-recorder for Rs. 1040 a man gains 4%. If he sells it for Rs. 950, his loss will be

167. Two trains 165 m and 135 m long run at the speed of 70 km/hr and 38 km/hr respectively in opposite directions on parallel tracks. The time (in seconds) which they take to cross each other is:

168. Two trains are moving in opposite directions at 70 kmph and at 80 kmph. Their lengths are 1.3 km and 0.7 km respectively. The time taken by the slower train to cross the faster train in seconds is:

169. A train 75 m long passes a person, running at 6 kmph in the same direction in which the train is going in 15 seconds. The speed of the train is:

170. A train 105 m long passes a person, running at 2 kmph in the direction opposite to that of the train, in 2 seconds. The speed of the train is:

171. A train 150 metres long crosses a milestone in 15 seconds and crosses another train of the same length travelling in the opposite direction in 12 seconds. The speed of the second train in km/hr

172. Two trains, A and B start from stations X and Y towards Y and X respectively. After passing each other, They take 4 hours 48 minutes and 3 hours 20 minutes to reach Y and X respectively. If train A is moving at 45 km/hr., then the speed of the train B is

173. A train covers a distance between station A and station B in 45 minutes. If the speed of the train is reduced by 5 km per hour, then the same distance is covered in 48 minutes. The distance between stations A and B is

174. A train 300 m long is running at a speed of 25 metres per second, it will cross a bridge 200 metres long in ?

175. A train passes by a lamp post on a platform in 7 sec. and passes by the platform completely in 28 sec. if the length of the platform is 390 m. then length of the train (in metres) is

176. Two trains move from station A and station B towards each other at the speed of 50 km/h and 60 km/h. At the meeting point, the driver of the second train felt that the train has covered 120 km more. What is the distance between A and B?

177. A boat takes 19 hours for travelling downstream from point A to Point B and coming back to a Point C midway between A and B. If the velocity of the stream is 4 kmph and the speed of the boat in still water is 12 kmph, what is the distance between A and B? A. 160 km B. 152 km C. 200 km D. 220 km

178. Speed of a boat in still water is 8 kmph and speed of stream is 1.5 kmph. A man rows to a place at a distance of 61.75 km and come back to starting point. The total time taken by him. A. 6 hrs B. 8 hrs C. 16 hrs D. 22 hrs

179. In stream running at 2 kmph, a motorboat goes 6 km upstream and back again to the starting point in 33 minutes. Find the speed of the motorboat in still water.

180. If a man rows at the rate of 5 kmph in still water and his rate against the current is 3.5 kmph. then the man's rate along the current is:

181. A man can row 18 kmph in still water. It takes him thrice as long to row up as to row down the river. Find the rate of stream.

182. A steamer takes 36 minutes less to travel 36 miles downstream than to travel the same distance upstream. If the speed of the steamer in still water is 6 miles per minute the speed of the stream (miles per minute) is: .

183. A man's downstream speed is 7 kmph and speed of current is 1 kmph, the upstream speed of the man is-

184. A boat travel with a speed of 10 kmph in still water. If the speed of the stream is 3 kmph then find time taken by boat to travel 52 km downstream.

185. A man can row upstream 12 km and 30 km downstream each in 6 hrs. the velocity (in kmph) of the current is:

186. A man takes 3 hours 45 minutes to row a boat 22.5 km downstream of a river and 2 hours 30 minutes to cover a distance of 10 km up stream. Find the speed of the river current in km/hr.

187. Find compound interest on Rs. 7000 at 21% per annum for 2 years 4 months, compounded annually.

188. Find the compound interest on Rs. 7500 in 2 years at 6% per annum, the interest being compounded half-yearly.

189. Find the compound interest on Rs. 10,000 at 20% per annum for 6 months. compounded quarterly.

190. If the simple interest on a sum of money at 5% per annum for 2 years is Rs. 1400, find the compound interest on the same sum for the same period at the same rate.

191. If Rs. 1000 amounts to Rs. 1166.40 in two years compounded annually, Find the rate of interest per annum.

192. $17.92 \times 10.97 - 52.11 = ?$ A. 146 B. 196 C. 216 D. 228 E. 91

193. $127.93 \times 4.1 - 116.01 \times 2.95 = ?$

194. $101.68 \div 5.93 = ? \div 11.01$.

195. $167.71 \div 6.09 \times 4.11 = 79.86\%$ of ?

196 59.71% of $(207.93 + 218.09) = (?)^2$

197. $(44.95\% \text{ of } 1200) + (65.06\% \text{ of } 800) = ?$

198. $11.231 + 35.891 + 53.756 - 58.111 = ?$

199. $(18.04 \times 19.88) \div 6.97 = (?)^2$

200. The numerator of a fraction is decreased by 50% and the denominator is increased by 200%. If the resultant fraction is $\frac{5}{2}$, find the $\frac{3}{20}$ th of the original fraction.

